

矩阵分析与应用

第十二讲 矩阵分解之二

本讲主要内容

- 矩阵的QR分解
- 矩阵的满秩分解
- 矩阵的奇异值分解

2. G-变换方法

定理8: $A_{n \times n}$ 可逆 $\Rightarrow \exists$ 有限个G-矩阵之积 T , 使得 TA 为可逆上三角矩阵。

证明: 以 $n=4$ 为例

$$|A| \neq 0 \quad \beta^{(0)} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix} \neq 0$$

$\Rightarrow \exists$ 有限个G-矩阵之积 T_0 , 使得

$$T_0 \beta^{(0)} = \begin{bmatrix} |\beta^{(0)}| \\ 0 \\ 0 \\ 0 \end{bmatrix}, a_{11}^{(1)} = |\beta^{(0)}| > 0$$

$$T_0 A = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & a_{14}^{(1)} \\ 0 & a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ 0 & a_{32}^{(1)} & a_{33}^{(1)} & a_{34}^{(1)} \\ 0 & a_{42}^{(1)} & a_{43}^{(1)} & a_{44}^{(1)} \end{bmatrix} \quad \begin{bmatrix} a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ a_{32}^{(1)} & a_{33}^{(1)} & a_{34}^{(1)} \\ a_{42}^{(1)} & a_{43}^{(1)} & a_{44}^{(1)} \end{bmatrix} = A^{(1)}$$

$$|A^{(1)}| \neq 0, \beta^{(1)} = \begin{bmatrix} a_{22}^{(1)} \\ a_{32}^{(1)} \\ a_{42}^{(1)} \end{bmatrix} \neq 0 \Rightarrow \exists \text{有限个G-矩阵之积 } T_1, \text{使得}$$

$$T_1 \beta^{(1)} = \begin{bmatrix} |\beta^{(1)}| \\ 0 \\ 0 \end{bmatrix}, a_{22}^{(2)} = |\beta^{(1)}| > 0$$

$$T_1 A^{(1)} = \begin{bmatrix} a_{22}^{(2)} & a_{23}^{(2)} & a_{24}^{(2)} \\ \mathbf{0} & a_{33}^{(2)} & a_{34}^{(2)} \\ \mathbf{0} & a_{43}^{(2)} & a_{44}^{(2)} \end{bmatrix}, \quad A^{(2)} = \begin{bmatrix} a_{33}^{(2)} & a_{34}^{(2)} \\ a_{43}^{(2)} & a_{44}^{(2)} \end{bmatrix}$$

$$|A^{(2)}| \neq \mathbf{0}, \beta^{(2)} = \begin{bmatrix} a_{33}^{(2)} \\ a_{43}^{(2)} \end{bmatrix} \neq \mathbf{0} \Rightarrow \exists \text{G-矩阵 } T_2, \text{使得}$$

$$T_2 \beta^{(2)} = \begin{bmatrix} |\beta^{(2)}| \\ \mathbf{0} \end{bmatrix}, a_{33}^{(3)} = |\beta^{(2)}| > \mathbf{0} \quad T_2 A^{(2)} = \begin{bmatrix} a_{33}^{(3)} & a_{34}^{(3)} \\ \mathbf{0} & a_{44}^{(3)} \end{bmatrix}$$

$$T = \begin{bmatrix} I_2 & \\ & T_2 \end{bmatrix} \begin{bmatrix} \mathbf{1} & \\ & T_1 \end{bmatrix} T_0 \quad TA = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & a_{14}^{(1)} \\ & a_{22}^{(2)} & a_{23}^{(2)} & a_{24}^{(2)} \\ & & a_{33}^{(3)} & a_{34}^{(3)} \\ & & & a_{44}^{(3)} \end{bmatrix} \triangleq R$$

2. G-变换方法

定理8: $A_{n \times n}$ 可逆 $\Rightarrow \exists$ 有限个G-矩阵之积 T , 使得 TA 为可逆上三角矩阵。

[注]: $\det T = 1 \Rightarrow \det A = a_{11}^{(1)} a_{22}^{(2)} \cdots a_{n-1,n-1}^{(n-1)} a_{n,n}^{(n-1)}$

因此 $a_{n,n}^{(n-1)}$ 与 $\det A$ 同符号

当 $A_{n,n}$ 不可逆时, 仍可得 $TA = R$, 但 R 是不可逆矩阵

例 5: 用G变换法求 $A = \begin{bmatrix} 3 & 5 & 5 \\ 0 & 3 & 4 \\ 4 & 0 & 5 \end{bmatrix}$, 的QR分解

解 (1) $\beta^{(0)} = \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix}$: $T_{13}(c, s)$ 中 $c = \frac{3}{5}, s = \frac{4}{5}$. $T_{13}\beta^{(0)} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$

$$T_0 \stackrel{\Delta}{=} T_{13} = \begin{bmatrix} 3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & 3/5 \end{bmatrix}, \quad T_0 A = \begin{bmatrix} 5 & 3 & 7 \\ 0 & 3 & 4 \\ 0 & -4 & -1 \end{bmatrix}$$

(2) $A^{(1)} = \begin{bmatrix} 3 & 4 \\ -4 & -1 \end{bmatrix}, \beta^{(1)} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$

$$T_{12}(c, s) \text{ 中 } c = \frac{3}{5}, s = -\frac{4}{5}. \quad T_{12}\beta^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$T_1 \triangleq T_{12} = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}, \quad T_1 A^{(1)} = \begin{bmatrix} 5 & 16/5 \\ 0 & 13/5 \end{bmatrix}$$

$$\text{令 } T = \begin{bmatrix} \mathbf{1} & \\ & T_1 \end{bmatrix} T_0 = \frac{1}{5} \begin{bmatrix} 5 & & \\ & 3 & -4 \\ & 4 & 3 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 & 0 & 4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 15 & 0 & 20 \\ 16 & 15 & -12 \\ -12 & 20 & 9 \end{bmatrix}$$

$$\text{则 } Q = T^{-1} = T^T, \quad R = \begin{bmatrix} 5 & 3 & 7 \\ & 5 & 16/5 \\ & & 13/5 \end{bmatrix}: \quad A = QR$$

例 6 用 Givens 变换求 $A = \begin{bmatrix} 0 & 0 & 1 & 3 \\ 0 & 4 & 2 & 3 \\ 1 & 1 & 2 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}$ 的 QR 分解.

(1) $\beta^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, 构造 $T_{13}(c, s)$, $c = 0, s = 1$, 则

$$T_{13}\beta^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad T_0 = T_{13} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad T_0 A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 4 & 3 & 2 \\ 0 & 0 & -1 & -3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

(2) $A^{(2)} = \begin{bmatrix} -1 & -3 \\ 1 & 2 \end{bmatrix}$, $\beta^{(2)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, 构造 $T_{12}(c, s)$, $c = \frac{-1}{\sqrt{2}}$, $s = \frac{1}{\sqrt{2}}$, 则

$$T_{12}\beta^{(2)} = \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix}, \quad T_2 = T_{12} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}, \quad T_2 A^{(2)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 2 & 5 \\ 0 & 1 \end{bmatrix}$$

$$\text{令 } T = \begin{bmatrix} I_2 & O \\ O & T_2 \end{bmatrix} T_0 = \frac{1}{\sqrt{2}} \left[\begin{array}{cc|cc} \sqrt{2} & 0 & & \\ 0 & \sqrt{2} & & \\ \hline & & -1 & 1 \\ & & -1 & -1 \end{array} \right] \cdot \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & \sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \end{bmatrix}$$

$$\text{则 } Q = T^T = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & \sqrt{2} & 0 & 0 \\ \sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 1 & 2 & 4 \\ & 4 & 3 & 2 \\ & & \sqrt{2} & 5/\sqrt{2} \\ & & & 1/\sqrt{2} \end{bmatrix}$$

$$A = QR$$

3. H-变换方法

定理10: $A_{n \times n}$ 可逆 $\Rightarrow \exists$ 有限个H-矩阵之积 S ,

使得 SA 为可逆上三角矩阵。

证明: 以 $n=4$ 为例

$$(1) |A| \neq 0: \beta^{(0)} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix} \neq 0 \Rightarrow \exists \text{H-矩阵 } H_0, \text{ 使得 } H_0 \beta^{(0)} = \begin{bmatrix} |\beta^{(0)}| \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_{11}^{(1)} = |\beta^{(0)}| > 0 \quad H_0 A = \left[\begin{array}{c|ccc} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & a_{14}^{(1)} \\ \hline 0 & & & \\ 0 & & A^{(1)} & \\ 0 & & & \end{array} \right], \quad A^{(1)} = \begin{bmatrix} a_{22}^{(1)} & a_{23}^{(1)} & a_{24}^{(1)} \\ a_{32}^{(1)} & a_{33}^{(1)} & a_{34}^{(1)} \\ a_{42}^{(1)} & a_{43}^{(1)} & a_{44}^{(1)} \end{bmatrix}$$

$$(2) \quad |A^{(1)}| \neq 0: \quad \beta^{(1)} = \begin{bmatrix} a_{22}^{(1)} \\ a_{32}^{(1)} \\ a_{42}^{(1)} \end{bmatrix} \neq 0 \Rightarrow \exists \mathbf{H}\text{-矩阵 } H_1, \text{ 使得 } H_1 \beta^{(1)} = \begin{bmatrix} |\beta^{(1)}| \\ 0 \\ 0 \end{bmatrix}$$

$$a_{22}^{(2)} = |\beta^{(1)}| > 0 \quad H_1 A^{(1)} = \begin{bmatrix} a_{22}^{(2)} & a_{23}^{(2)} & a_{24}^{(2)} \\ 0 & a_{33}^{(2)} & a_{34}^{(2)} \\ 0 & a_{43}^{(2)} & a_{44}^{(2)} \end{bmatrix}, \quad A^{(2)} = \begin{bmatrix} a_{33}^{(2)} & a_{34}^{(2)} \\ a_{43}^{(2)} & a_{44}^{(2)} \end{bmatrix}$$

$$(3) \quad |A^{(2)}| \neq 0: \quad \beta^{(2)} = \begin{bmatrix} a_{33}^{(2)} \\ a_{43}^{(2)} \end{bmatrix} \neq 0 \Rightarrow \exists \mathbf{H}\text{-矩阵 } H_2, \text{ 使得 } H_2 \beta^{(2)} = \begin{bmatrix} |\beta^{(2)}| \\ 0 \end{bmatrix}$$

$$a_{33}^{(3)} = |\beta^{(2)}| > 0, \quad H_2 A^{(2)} = \begin{bmatrix} a_{33}^{(3)} & a_{34}^{(3)} \\ 0 & a_{44}^{(3)} \end{bmatrix}$$

$$\text{令 } S = \begin{bmatrix} I_2 & \\ & H_2 \end{bmatrix} \cdot \begin{bmatrix} 1 & \\ & H_1 \end{bmatrix} \cdot H_0 \quad SA = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & a_{14}^{(1)} \\ & a_{22}^{(2)} & a_{23}^{(2)} & a_{24}^{(2)} \\ & & a_{33}^{(3)} & a_{34}^{(3)} \\ & & & a_{44}^{(3)} \end{bmatrix} \triangleq R$$

例 7 用 H-变换求 $A = \begin{bmatrix} 3 & 14 & 9 \\ 6 & 43 & 3 \\ 6 & 22 & 15 \end{bmatrix}$ 的 QR 分解

解 (1) $\beta^{(0)} = \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}$, $\beta^{(0)} - |\beta^{(0)}|e_1 = \begin{bmatrix} -6 \\ 6 \\ 6 \end{bmatrix} = 6 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, $u = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$$H_0 = I - 2uu^T = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}, \quad H_0 A = \begin{bmatrix} 9 & 48 & 15 \\ 0 & 9 & -3 \\ 0 & -12 & 9 \end{bmatrix}$$

$$(2) \quad A^{(1)} = \begin{bmatrix} 9 & -3 \\ -12 & 9 \end{bmatrix}, \quad \beta^{(1)} = \begin{bmatrix} 9 \\ -12 \end{bmatrix}$$

$$\beta^{(1)} - |\beta^{(1)}|e_1 = \begin{bmatrix} -6 \\ -12 \end{bmatrix} = (-6) \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad u = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$H_1 = I - 2uu^T = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix}, \quad H_1 A^{(1)} = \begin{bmatrix} 15 & -9 \\ 0 & -3 \end{bmatrix}$$

$$\text{令 } S = \begin{bmatrix} 1 & \\ & H_1 \end{bmatrix} H_0$$

$$\text{则 } Q = S^{-1} = S^T = H_0 \begin{bmatrix} 1 & \\ & H_1 \end{bmatrix} = \frac{1}{15} \begin{bmatrix} 5 & -2 & -14 \\ 10 & 11 & 2 \\ 10 & -10 & 5 \end{bmatrix}$$

$$R = \begin{bmatrix} 9 & 48 & 15 \\ & 15 & -9 \\ & & -3 \end{bmatrix} : A = QR$$

五、化方阵与Hessenberg矩阵相似

上 Hessenberg 矩阵: $F_{\text{上}} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ & \ddots & \ddots & \ddots & \vdots \\ & & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} \\ & & & a_{n,n-1} & a_{nn} \end{bmatrix}$

定理11: $A_{n \times n}$, 则存在有限个**G**-矩阵之积**Q**, 使得

$$QAQ^T = F_{\text{上}}$$

定理12: $A_{n \times n}$, 则存在有限个**H**-矩阵之积**Q**, 使得

$$QAQ^T = F_{\text{上}}$$

推论: $A_{n \times n}$ 实对称 $\Rightarrow \exists$ 存在有限个**H**-矩阵(**G**-矩阵)之积**Q**, 使得 $QAQ^T =$ “实对称三对角矩阵”

定理11: $A_{n \times n}$, 则存在有限个**G**-矩阵之积**Q**, 使得

$$QAQ^T = F_{\text{上}}$$

证明: (1) 对A: 如果 $\beta^{(0)} = \begin{bmatrix} a_{21} \\ \vdots \\ a_{n1} \end{bmatrix} \neq \mathbf{0}$, 则存在有限个

G矩阵之积 T_0 , 使得 $T_0 \beta^{(0)} = |\beta^{(0)}| e_1 = a_{21}^{(1)} e_1$

$$\begin{bmatrix} \mathbf{1} & \\ & T_0 \end{bmatrix} A \begin{bmatrix} \mathbf{1} & \\ & T_0 \end{bmatrix}^T = \begin{bmatrix} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1n}^{(1)} \\ a_{21}^{(1)} & & & & \\ \mathbf{0} & & & & \\ \vdots & & & & \\ \mathbf{0} & & & & \end{bmatrix} A^{(1)}$$

如果 $\beta^{(0)} = \mathbf{0}$, 转入 (2)

(2) 对 $\mathbf{A}^{(1)}$: 如果 $\boldsymbol{\beta}^{(1)} = \begin{bmatrix} a_{32}^{(1)} \\ \vdots \\ a_{n2}^{(1)} \end{bmatrix} \neq \mathbf{0}$, 则存在有限个G矩阵

之积 $\mathbf{T}_1$, 使得 $\mathbf{T}_1 \boldsymbol{\beta}^{(1)} = |\boldsymbol{\beta}^{(1)}| \mathbf{e}_1 = a_{32}^{(2)} \mathbf{e}_1$

$$\begin{bmatrix} \mathbf{1} & \mathbf{T}_1 \end{bmatrix} \mathbf{A}^{(1)} \begin{bmatrix} \mathbf{1} \\ \mathbf{T}_1 \end{bmatrix}^T = \begin{bmatrix} a_{22}^{(2)} & a_{23}^{(2)} & a_{24}^{(2)} & \cdots & a_{2n}^{(2)} \\ a_{32}^{(2)} & & & & \\ \mathbf{0} & & & & \\ \vdots & & & & \\ \mathbf{0} & & & & \end{bmatrix} \quad \mathbf{A}^{(2)}$$

如果 $\boldsymbol{\beta}^{(1)} = \mathbf{0}$, 转入 (3)

(3) 对 $A^{(2)}$:, 直到 $n-2$ 步结束

$$\text{令 } Q = \begin{bmatrix} I_{n-2} & \\ & T_{n-3} \end{bmatrix} \cdots \begin{bmatrix} I_2 & \\ & T_1 \end{bmatrix} \begin{bmatrix} 1 & \\ & T_0 \end{bmatrix}$$

$$\text{则 } QAQ^T = F_{\text{上}}$$

例8: 用H-变换化 $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ 正交相似于“三对角矩阵”

解: $\beta^{(0)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} : \beta^{(0)} - |\beta^{(0)}| e_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, u = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$$H_0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & & \\ & H_0 & \\ & & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$QA = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad QAQ^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

满秩分解

目的：对 $A \in C_r^{m \times n} (n \geq 1)$, 求 $F \in C_r^{m \times r}$, 及 $G \in C_r^{r \times n}$ 使 $A = FG$

分解原理：

$$\text{rank} A = r \Rightarrow A \xrightarrow{\text{行}} \text{梯形形 } B = \begin{pmatrix} G \\ O \end{pmatrix} : G \in C_r^{r \times n}$$

$$\Rightarrow \exists \text{有限个初等矩阵之积 } P_{m \times m}, \text{st. } PA = B$$

$$\Rightarrow A = P^{-1}B = \left(F_{m \times r} \mid S_{m \times (m-r)} \right) \begin{pmatrix} G \\ O \end{pmatrix} = FG : F \in C_r^{m \times r}$$

例9: $A = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 2 & -1 & 1 \\ 2 & 2 & -2 & -1 \end{bmatrix}$, 求 $A=FG$

解 (1) $(A|I) = \left[\begin{array}{cccc|ccc} -1 & 0 & 1 & 2 & 1 & & \\ 1 & 2 & -1 & 1 & & 1 & \\ 2 & 2 & -2 & -1 & & & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} -1 & 0 & 1 & 2 & 1 & & \\ 0 & 2 & 0 & 3 & 1 & 1 & \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}, P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix}, F = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 0 & 2 & 0 & 3 \end{bmatrix} \quad \text{满秩分解为 } A=FG$$

例9: $A = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 2 & -1 & 1 \\ 2 & 2 & -2 & -1 \end{bmatrix}$, 求 $A=FG$

解 (2) $(A|I) = \left[\begin{array}{cccc|cc} -1 & 0 & 1 & 2 & 1 & \\ 1 & 2 & -1 & 1 & & 1 \\ 2 & 2 & -2 & -1 & & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|ccc} 1 & 0 & -1 & -2 & -1 & & \\ 0 & 1 & 0 & 3/2 & 1/2 & 1/2 & \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]$

$$A = P^{-1}B = (F | S) \begin{pmatrix} I_2 & B_{12} \\ O & O \end{pmatrix} = (F | FB_{12})$$

故 $F = \text{“}A \text{ 的前 2 列”} = \begin{bmatrix} -1 & 0 \\ 1 & 2 \\ 2 & 2 \end{bmatrix}$, $G = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 0 & 3/2 \end{bmatrix}$

奇异值分解(SVD)

一、预备知识

(1) $\forall A_{m \times n}$, $(A^H A)_{n \times n}$ 是 **Hermite** (半) 正定矩阵.

$$\forall x \neq 0, x^H A^H A x = (Ax)^H (Ax) = |Ax|^2 \geq 0$$

(2) 齐次方程组 $Ax = 0$ 与 $A^H Ax = 0$ 同解

若 $Ax = 0$, 则 $A^H Ax = 0$;

反之, $A^H Ax = 0 \Rightarrow |Ax|^2 = (Ax)^H (Ax) = x^H (A^H Ax) = 0$
 $\Rightarrow Ax = 0$

$$(3) \quad \mathbf{rank} \, A = \mathbf{rank}(A^{\mathbf{H}} A)$$

$$S_1 = \{x \mid Ax = 0\}, \quad S_2 = \{x \mid A^{\mathbf{H}} Ax = 0\}$$

$$S_1 = S_2 \Rightarrow \dim S_1 = \dim S_2 \Rightarrow n - r_A = n - r_{A^{\mathbf{H}} A}$$

$$\Rightarrow r_A = r_{A^{\mathbf{H}} A}$$

$$(4) \quad A = \mathbf{O}_{m \times n} \Leftrightarrow A^{\mathbf{H}} A = \mathbf{O}_{n \times n}$$

必要性. 左乘 $A^{\mathbf{H}}$ 即得;

$$\text{充分性} \quad r_A = r_{A^{\mathbf{H}} A} = 0 \Rightarrow A = \mathbf{O}$$

二、正交对角分解

定理15: $A_{n \times n}$ 可逆 $\Rightarrow \exists$ 酉矩阵 $U_{n \times n}, V_{n \times n}$, 使得

$$U^H A V = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{bmatrix} \triangleq D \quad (\sigma_i > 0)$$

证: $A^H A$ 是 **Hermite** 正定矩阵, $\exists$ 酉矩阵 $V_{n \times n}$ 使得

$$V^H (A^H A) V = \text{diag}(\lambda_1, \dots, \lambda_n) \triangleq \Lambda \quad (\lambda_i > 0)$$

改写为 $D^{-1} V^H A^H \cdot A V D^{-1} = I \quad (\sigma_i = \sqrt{\lambda_i})$

令 $U = A V D^{-1}$, 则有 $U^H U = I$, 从而 U 是酉矩阵。

由此可得 $U^H A V = U^H U D = D$

三、奇异值分解

$$A_{m \times n} \in C_r^{m \times n} (r \geq 1) \Rightarrow A^H A \in C_r^{n \times n} \text{ 半正定}$$

$$A^H A \text{ 的特征值: } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r \geq \lambda_{r+1} = \dots = \lambda_n = 0$$

$$A \text{ 的奇异值: } \sigma_i = \sqrt{\lambda_i}, \quad i = 1, 2, \dots, n$$

特点：（1） A 的奇异值个数等于 A 的列数

（2） A 的非零奇异值个数等于 $\text{rank } A$

定理16: $A_{m \times n} \in C_r^{m \times n} (r \geq 1), \Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$

存在酉矩阵 $U_{m \times m}$ 及 $V_{n \times n}$, 使得 $U^H A V = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n} \triangleq D$

证明: 对于Hermite半正定矩阵 $A^H A$, 存在酉矩阵 $V_{n \times n}$

$$V^H (A^H A) V = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = \begin{bmatrix} \Sigma^2 & 0 \\ 0 & 0 \end{bmatrix}_{n \times n} \quad \sigma_i = \sqrt{\lambda_i}, \quad i = 1, \dots, r$$

划分 $V = [V_1 \ V_2]$, V_1 是 V 的前 r 列, V_2 是后 $n-r$ 列

$$(A^H A) V = V \begin{bmatrix} \Sigma^2 & 0 \\ 0 & 0 \end{bmatrix}_{n \times n} \Rightarrow [(A^H A) V_1 \mid (A^H A) V_2] = [V_1 \Sigma^2 \mid 0]$$

$$\begin{aligned} (1) \quad (A^H A) V_1 &= V_1 \Sigma^2 & \Rightarrow V_1^H (A^H A) V_1 &= \Sigma^2 \\ \Rightarrow \Sigma_r^{-1} V_1^H A^H A V_1 \Sigma_r^{-1} &= I_r & \Rightarrow (A V_1 \Sigma^{-1})^H (A V_1 \Sigma^{-1}) &= I_r \end{aligned}$$

$$(2) \quad (A^H A)V_2 = 0 \quad \Rightarrow V_2^H (A^H A)V_2 = 0$$

$$\Rightarrow (AV_2)^H (AV_2) = 0 \quad \Rightarrow (AV_2) = 0$$

$$\text{令 } U_1 \triangleq AV_1 \Sigma^{-1}, \text{ 有 } U_1^H U_1 = I_r \quad \because (AV_1 \Sigma^{-1})^H (AV_1 \Sigma^{-1}) = I_r$$

设 U_1 的列为 $u_1, \dots, u_r$, 扩充为 C^m 的基 $u_1, \dots, u_r, \mathbf{u}_{r+1}, \dots, \mathbf{u}_m$

则 $U_2 = (\mathbf{u}_{r+1}, \dots, \mathbf{u}_m)$ 满足 $U_2^H U_1 = 0_{(m-r) \times r}$ 。记 $U = (U_1 \ U_2)$

$$U^H AV = U^H (AV_1 \ AV_2) = \begin{bmatrix} U_1^H \\ U_2^H \end{bmatrix} (U_1 \Sigma_r \ 0)$$

$$= \begin{bmatrix} U_1^H U_1 \Sigma_r & 0 \\ U_2^H U_1 \Sigma_r & 0 \end{bmatrix} = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$$

定理16: $A_{m \times n} \in C_r^{m \times n} (r \geq 1), \Sigma_r = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix},$

存在酉矩阵 $U_{m \times m}$ 及 $V_{n \times n}$, 使得 $U^H A V = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}_{m \times n} \triangleq D$

[注]: 称 $A = U D V^H$ 为 A 的奇异值分解

- (1) U 与 V 不唯一;
- (2) U 的列为 $A A^H$ 的特征向量, V 的列为 $A^H A$ 的特征向量
- (3) 称 U 的列为 A 的左奇异向量, 称 V 的列为 A 的右奇异向量.

例10: 称 $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, 求 $A = UDV^T$

解: $AA^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} = B, |\lambda I - B| = \lambda(\lambda - 1)(\lambda - 3)$

$$\lambda_1 = 3: 3I - B = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & -1 \\ -1 & -1 & 1 \end{bmatrix}, \xi_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\lambda_2 = 1: 1I - B = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \xi_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\lambda_3 = 0 : \quad 0I - B = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \\ -1 & -1 & -2 \end{bmatrix}, \xi_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$r_A = 2 : \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{bmatrix}, \quad V_1 = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & 0 \end{bmatrix}$$

$$U_1 = AV_1\Sigma^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & 0 \end{bmatrix}, \text{ 取 } U_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \text{ 则 } U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$U^T AV = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = D, \quad A = UDV^T$$

定理17: $A \in C_r^{m \times n} (r \geq 0)$ 的奇异值分解 $A = U \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V^H$

中, 划分 $U = (u_1, u_2, \dots, u_m), V = (v_1, v_2, \dots, v_n)$, 则有

$$(1) \quad N(A) = \text{span}\{v_{r+1}, v_{r+2}, \dots, v_n\};$$

$$(2) \quad R(A) = \text{span}\{u_1, u_2, \dots, u_r\};$$

$$(3) \quad A = \sigma_1 u_1 v_1^H + \sigma_2 u_2 v_2^H + \dots + \sigma_r u_r v_r^H$$

证明:
$$A = (U_1 | U_2) \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_1^H \\ V_2^H \end{pmatrix} = U_1 \Sigma V_1^H$$

容易验证:
$$U_1 \Sigma V_1^H x = 0 \Leftrightarrow V_1^H x = 0$$

$$\begin{aligned}
(1) \quad N(A) &= \{ \mathbf{x} \mid A\mathbf{x} = \mathbf{0} \} = \{ \mathbf{x} \mid U_1 \Sigma V_1^H \mathbf{x} = \mathbf{0} \} \\
&= \{ \mathbf{x} \mid V_1^H \mathbf{x} = \mathbf{0} \} = N(V_1^H) = R^\perp(V_1) \\
&= R(V_2) = \text{span}\{ \mathbf{v}_{r+1}, \dots, \mathbf{v}_n \} \\
(2) \quad R(A) &= \{ \mathbf{y} \mid \mathbf{y} = A\mathbf{x} \} = \{ \mathbf{y} \mid \mathbf{y} = U_1 (\Sigma V_1^H \mathbf{x}) \} \\
&\subset \{ \mathbf{y} \mid \mathbf{y} = U_1 \mathbf{z} \} = R(U_1) \\
R(U_1) &= \{ \mathbf{y} \mid \mathbf{y} = U_1 \mathbf{z} \} = \{ \mathbf{y} \mid \mathbf{y} = A(V_1 \Sigma^{-1} \mathbf{z}) \} \\
&\subset \{ \mathbf{y} \mid \mathbf{y} = A\mathbf{x} \} = R(A) \\
\text{故} \quad R(A) &= R(U_1) = \text{span}\{ \mathbf{u}_1, \dots, \mathbf{u}_r \} \\
(3) \quad A &= (\mathbf{u}_1, \dots, \mathbf{u}_r) \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^H \\ \vdots \\ \mathbf{v}_r^H \end{bmatrix} \\
&= \sigma_1 \mathbf{u}_1 \mathbf{v}_1^H + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^H
\end{aligned}$$

四、正交相抵

$A_{m \times n}, B_{m \times n}$, 若有酉矩阵 $U_{m \times m}$ 及 $V_{n \times n}$, 使 $U^H A V = B$,
称 A 与 B 正交相抵。

性质: (1) A 与 A 正交相抵;

(2) A 与 B 正交相抵 $\Rightarrow B$ 与 A 正交相抵;

(3) A 与 B 正交相抵, B 与 C 正交相抵,

则 A 与 C 正交相抵

定理18: A 与 B 正交相抵 $\Rightarrow \sigma_A = \sigma_B$

证明: $B = U^H A V \Rightarrow B^H B = \dots = V^{-1} (A^H A) V$

$$\Rightarrow \lambda_{B^H B} = \lambda_{A^H A} \geq 0$$

$$\Rightarrow \sigma_A = \sigma_B$$

例: $A^H = A \Rightarrow \sigma_A = |\lambda_A|$ $\because \lambda_{A^H A} = \lambda_{A^2} = (\lambda_A)^2$

$A^H = -A \Rightarrow \sigma_A = |\lambda_A|$ $\because \lambda_{A^H A} = \lambda_{(jA)^2} = (j\lambda_A)^2$

$A^H = -A \Rightarrow \lambda_A$ 为0或纯虚数, $j\lambda_A$ 为实数

矩阵分解的应用

设方程组 $A_{m \times n} x = b$ 有解, 则有

(1) $m = n : A = LU \Rightarrow Ly = b, Ux = y$

(2) $m = n : A = QR \Rightarrow Rx = Q^T b$

(3) $A = UDV^H \Rightarrow Dy = U^H b \stackrel{\text{def}}{=} c, V^H x = y$

$$D = \begin{bmatrix} \Sigma & O \\ O & O \end{bmatrix}_{m \times n}, \quad \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}$$

$$\begin{bmatrix} \Sigma & O \\ O & O \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} \quad (\text{隐含 } c_{r+1} = 0, \dots, c_m = 0)$$

通解为

$$\begin{bmatrix} y_1 \\ \vdots \\ y_r \\ y_{r+1} \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} c_1/\sigma_1 \\ \vdots \\ c_r/\sigma_r \\ k_1 \\ \vdots \\ k_{n-r} \end{bmatrix} \quad (k_1, \dots, k_{n-r} \text{ 是任意常数})$$

$$\mathbf{x} = V \mathbf{y} = \left(\frac{c_1}{\sigma_1} \mathbf{v}_1 + \dots + \frac{c_r}{\sigma_r} \mathbf{v}_r \right) + (k_1 \mathbf{v}_{r+1} + \dots + k_{n-r} \mathbf{v}_n)$$

[注] $k_1 \mathbf{v}_{r+1} + \dots + k_{n-r} \mathbf{v}_n$ 是 $A_{m \times n} \mathbf{x} = \mathbf{0}$ 的通解

$$\text{因为 } A \left(\frac{c_1}{\sigma_1} \mathbf{v}_1 + \dots + \frac{c_r}{\sigma_r} \mathbf{v}_r \right) = A V_1 \Sigma^{-1} \begin{bmatrix} c_1 \\ \vdots \\ c_r \end{bmatrix} = U_1 \begin{bmatrix} c_1 \\ \vdots \\ c_r \end{bmatrix} = [U_1 \mid U_2] \mathbf{c} = \mathbf{b}$$

所以 $\frac{c_1}{\sigma_1} \mathbf{v}_1 + \dots + \frac{c_r}{\sigma_r} \mathbf{v}_r$ 是 $A_{m \times n} \mathbf{x} = \mathbf{b}$ 的一个特解

作业

- P195 1、 2、 3、 4
- P219 1、 2、 4 、 7
- P220 8、 9
- P225 1、 2、 3、 4
- P233 1、 2、 4